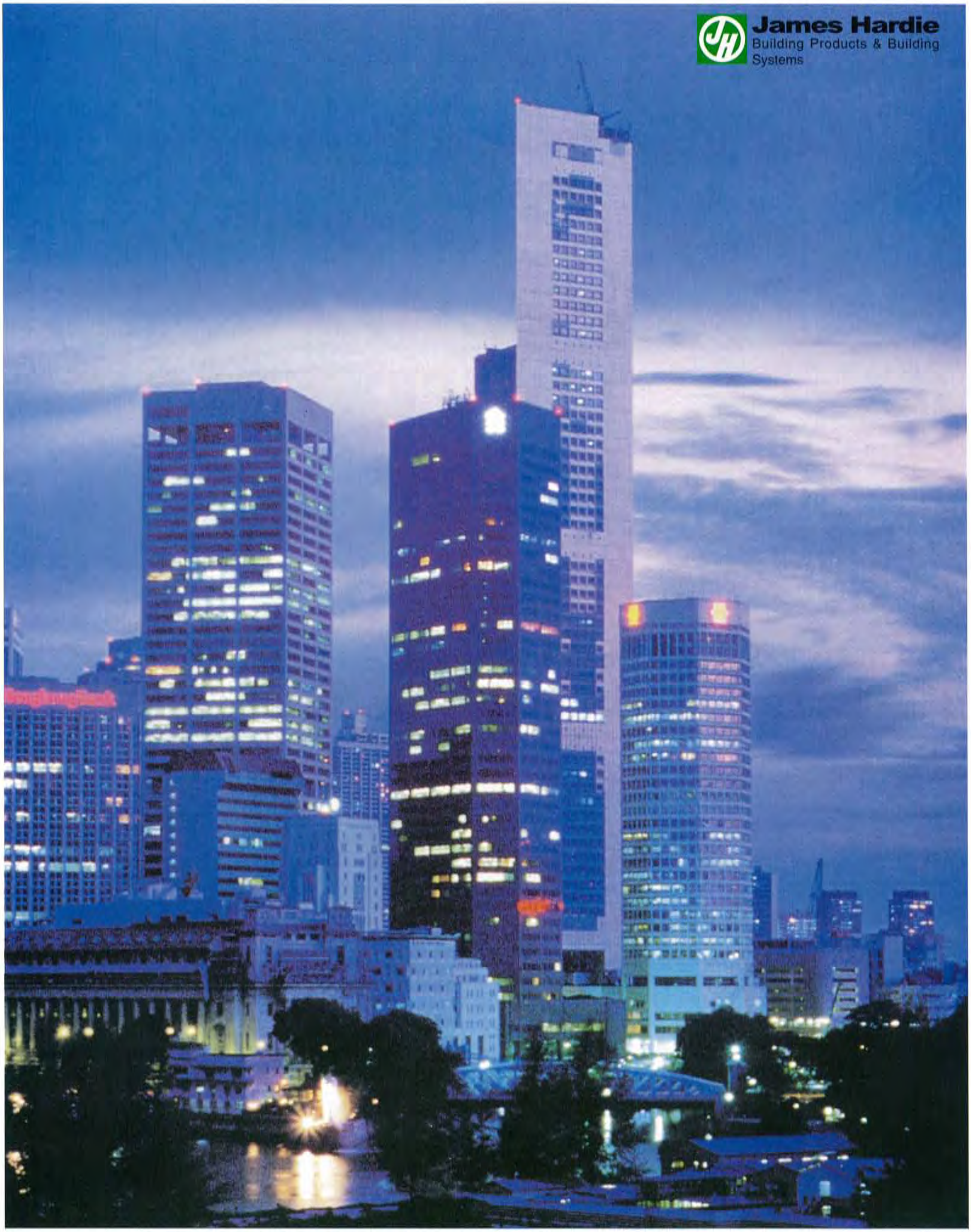
James Hardie Villaboard™

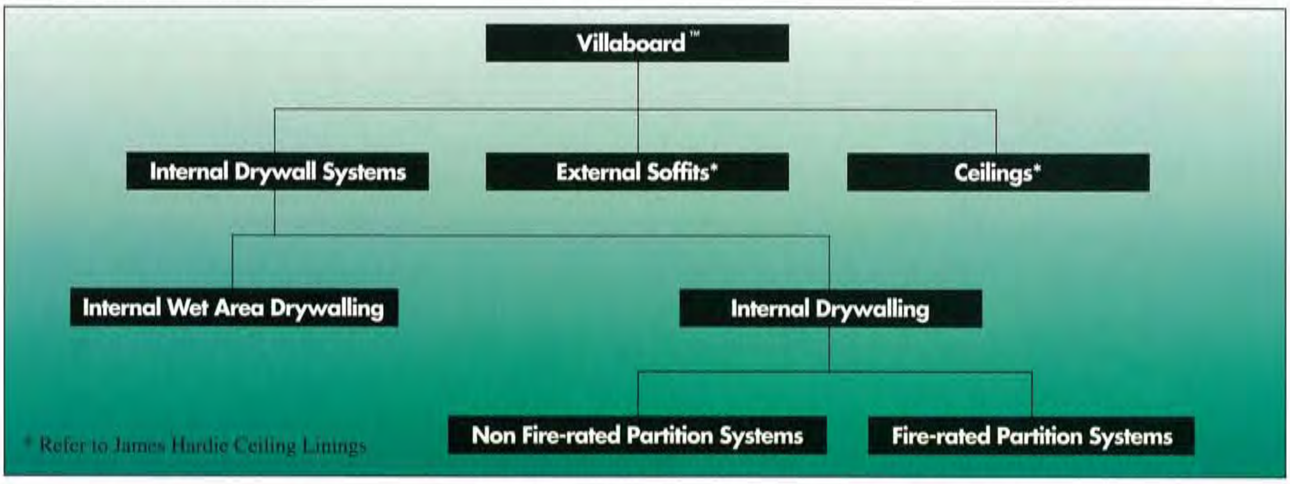
INTERNAL DRYWALL SYSTEMS



•LIGHTWEIGHT•ECONOMICAL•DURABLE•VERSATILE•

James Hardie Villaboard™

James Hardie Villaboard™ is a high quality, high impact resistant, internal lining board ideal for walls and ceilings, in both domestic and commercial construction. The versatility offered by Villaboard™, enables it to be confidently specified in any of the following applications:



Quality Composition

Villaboard™ is a high quality, versatile lining board, composed of Portland cement, ground sand, water (forming calcium silicate) and cellulose reinforcing fibre. It is manufactured exclusively by James Hardie & Coy Pty Limited, a leader in asbestos-free fibre cement technology.

Attractive Finishes

Villaboard™ is manufactured with recessed edges for flush jointing to allow a smooth joint-free finished surface. The finishing face is sanded for a smooth, scuff-resistant surface far superior than rendered masonry or lightweight concrete blockwork.

Durability and Proven Performance

Villaboard™ is completely immune to permanent water damage, does not rot and is unaffected by termites, sunlight or steam. Within normal applications, the product requires minimal maintenance and has a life limited only by the durability of the supporting structure and the workmanship of the installation.

The high impact strength and the hard wearing surface, give Villaboard™, distinct advantages over inferior paper-faced boards in both performance and long term durability aspects. It is entirely suitable for buildings such as schools, hospitals and in transit areas of airports.

Climate Effectiveness

Villaboard's capacity to resist long or short term exposure to moisture without degradation makes it an ideal substrate for ceramic or marble tiles in wet areas. Furthermore, it is ideal for use in humid climates, where paper-faced boards are deficient and have been prone to deterioration.

Versatility

Villaboard™, can be used throughout your building's interior as the total wall lining material, suitable for dry areas, wet areas, and ceilings.

Drywalling living areas

Internal Wet Area Drywalling

James Hardie Villaboard™

INTERNAL DRYWALL SYSTEMS

In the design of a modern building, internal walls must not only define the spaces within the building, but must meet additional performance requirements, which include:-

- Fire Resistance
- Acoustic Properties
- Moisture Resistance
- Lightweight
- Resistance to Impact & Surface Abrasion

In addition, a wall system must be easily and speedily constructed with the minimum of disturbance to other trades and be economically viable.

James Hardie Villaboard™ Drywall Systems will meet all these requirements.

Fire Resistant and Acoustically Rated Walls

James Hardie Fire Rated Drywall Partition Systems consisting of steel frames clad with composite linings of fire resistant grade Gypsum Wallboard and Villaboard™, offer excellent fire and acoustic performance. Fire resistance ratings of up to 2 hours and STC ratings of up to 60 can be readily achieved with different lining thicknesses and the use of acoustic foam in wall cavities.

Impact and Moisture Resistant

The excellent physical properties of Villaboard™, provides James Hardie Villaboard™ Drywall Systems with the unique benefit of both high impact and moisture resistance.

Lightweight for Fast Track Construction. Avoids Wet Trades.

James Hardie Villaboard™ Drywall Partition Systems, at less than a third the weight of masonry, offer many of the benefits associated with masonry and lightweight concrete block, with the added advantage of offering savings in both time and labour by avoiding the inconvenience of wet trades. Furthermore, unlike the permanency of masonry construction, James Hardie Villaboard™ Drywall partitions offer the convenience and flexibility of being fully relocatable.

The Company Behind The Products

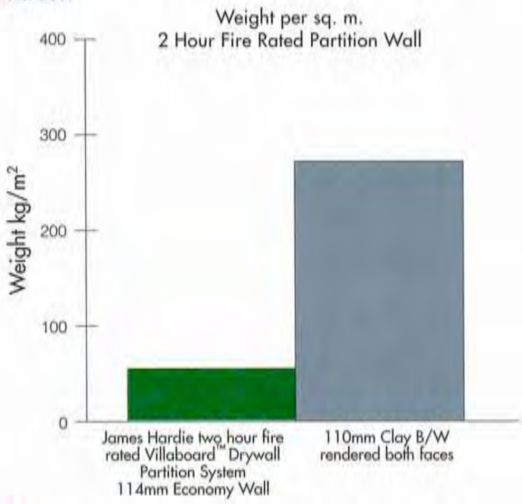
James Hardie & Coy Pty Limited is an Australian company which knows the building industry intimately. The company is a world leader in the development and manufacture of asbestos-free fibre cement building products, such as Hardipanel™ Compressed, Hardiplank™, Hardiflex™, Villaboard™, and Versilux™.
James Hardie Villaboard™ and James Hardie Villaboard™ Drywall Systems are products of modern fibre cement technology, that have gained a wide acceptance within the construction industry. These Systems have been tested and comply with relevant building codes and have been approved for use by Authorities throughout the world.

James Hardie Quality

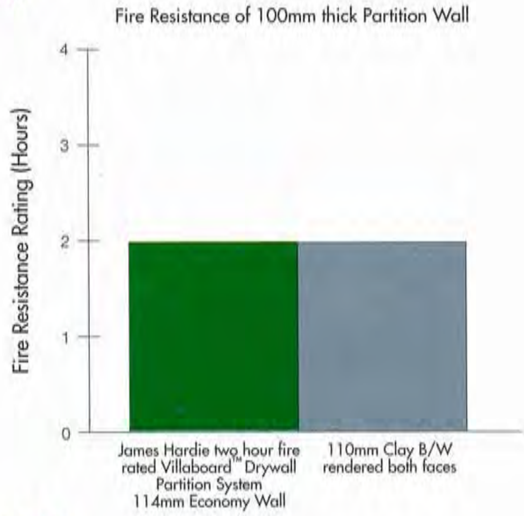
James Hardie is committed to achieving Total Quality Management. Currently, all James Hardie building products are manufactured under a system satisfying the International Accreditation System, ISO 9002, to guarantee the highest level of quality. This ensures that by specifying James Hardie, you are also specifying quality product, that will perform and confidently meet your expectations.

Performance Comparison

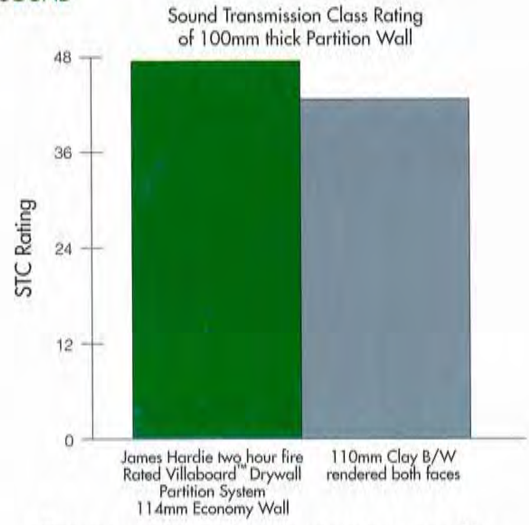
WEIGHT



FIRE



SOUND



For your construction requirements specify James Hardie Villaboard™ Drywall Systems! For further technical and sales information contact your local James Hardie representative.

James Hardie Villaboard™

The Philippines Stock Exchange

World Trade Centre, Singapore

Northpoint Shopping Centre, Singapore

The Hong Kong Football Club, Hong Kong

Bathroom in Albury Base Hospital, Australia

Modern bathroom in a private residence, Australia

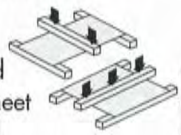
Commercial Internal Linings

Villaboard™ Physical Properties

Where values are quoted at Equilibrium Moisture Content (EMC) the conditions of the environment are nominally 23ºC and 50% relative humidity.

Density Average at EMC		1380 kg/ m^3
Mass at EMC:		
6mm thick		9 kg/ m^2
9mm thick		13 kg/ m^2
Moisture Content:		
at EMC		7%
at Saturation		33%
Moisture Movement:		
From EMC to Saturation		0.06%
From EMC to Oven Dry		0.11%
TOTAL		0.17%
Thermal Conductivity:		
at 40°C		0.47 W/ mK
Coeff. of Thermal Expansion		7.5 x 10^{-6} / K
Modules of Elasticity	Typical Values	
	at EMC (GPa)	Saturated (GPa)
Determined in static cross bending	6	4

Shear Strength (Estimated)		Typical Values	
		at EMC (MPa)	Saturated (MPa)
Perpendicular to the plane of the sheet		18	14
Compressive Strength (Estimated)		Typical Values	
		at EMC (MPa)	Saturated (MPa)
In the plane of the sheet		20	15
Perpendicular to the plane of the sheet		Better than 50	
Flexural Strength		Characteristic at EMC (MPa)	MOR[1] Saturated (MPa)
Line Load across sheet	Ultimate	18	11
	Proof[2]	16	8
Line load along the sheet	Ultimate	14	9
	Proof[2]	13	6



1. Characteristic MOR is that strength value, determined in bending, which is exceeded by not less than 95% of the product population.
2. "Proof" stress is that at which the Villaboard™, ceases to behave in a generally elastic manner. "Proof" stress may be likened to the "Yield" stress in steel for design purposes.

Sheet Dimensions							
Thickness (mm)	Widths (mm)	Lengths (mm)					
6.0	1200	3600	3000	2700	2400	2100	1800
	900	3600	-	-	2400	-	-
9.0	1200	-	-	2700	2400	-	1800
	900	-	-	-	2400	-	-

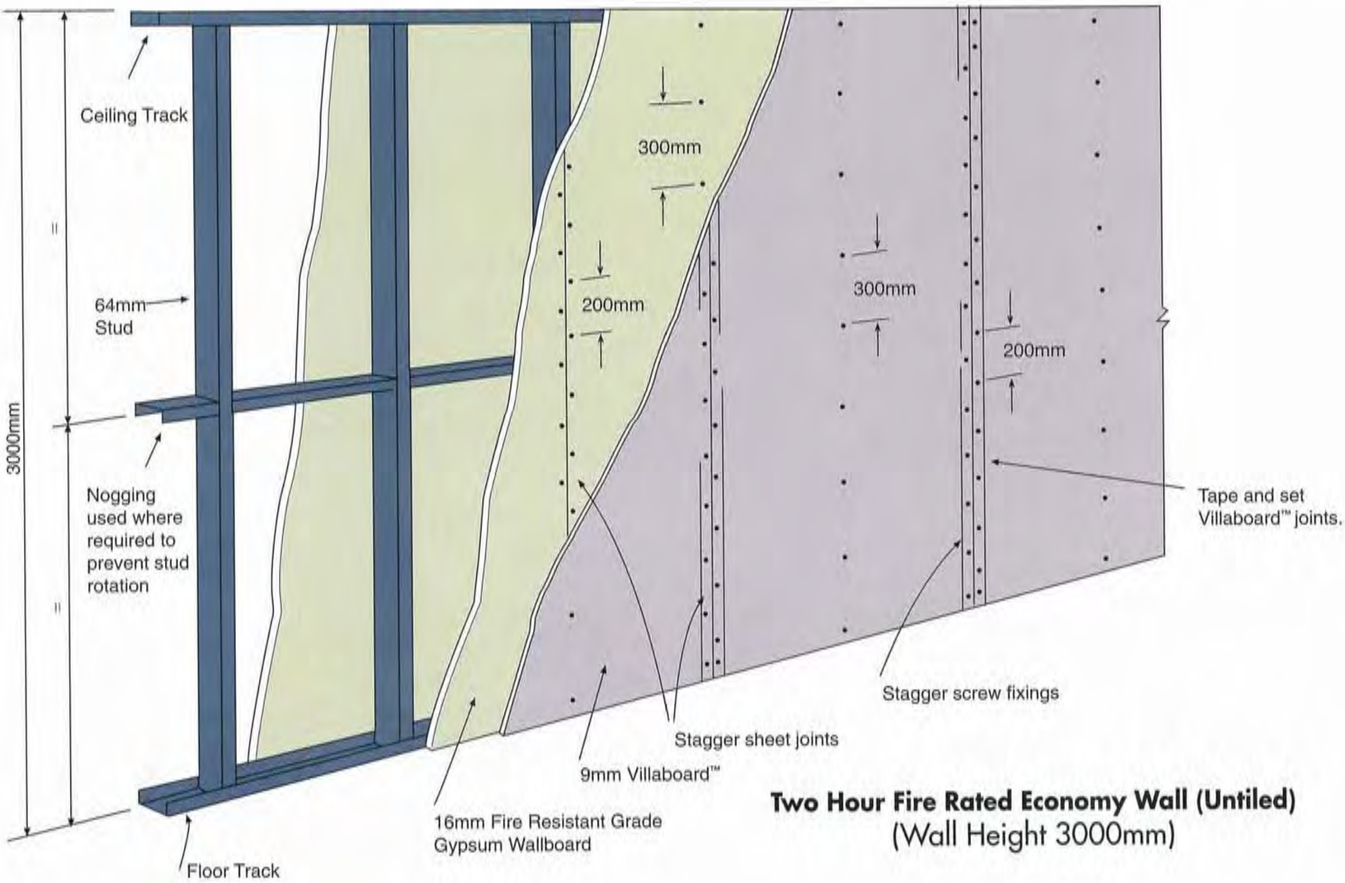
Note: Villaboard® is also available in 12 mm thickness and imperial dimensions to order.

CORPORATE HEALTH CENTRE, Auckland, New Zealand

100m^2 of 9mm Villaboard™ and 350m^2 of 6mm Villaboard™ used for tiled walls, shower enclosures and ceilings.

James Hardie Villaboard™

Fire-rated Drywall Partition System



Two Hour Fire Rated Economy Wall (Untiled)
(Wall Height 3000mm)

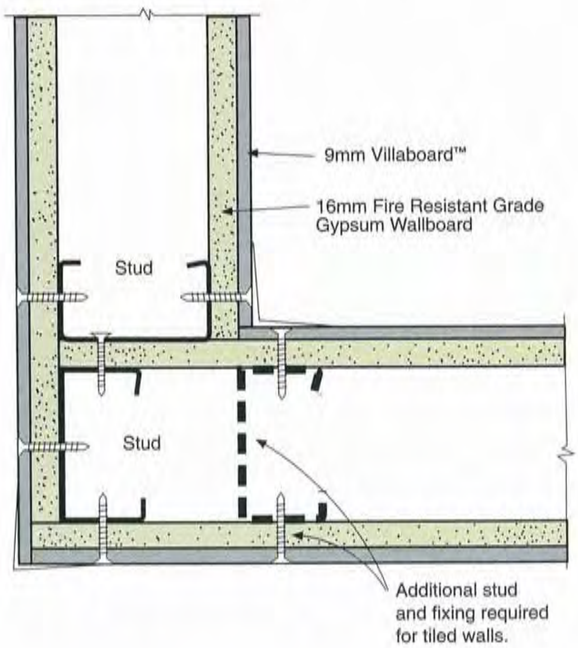
James Hardie Villaboard™ Fire-rated Drywall Partition System		
Fire Resistance	Acoustics	Constructions
1/2 Hour 30/30/30	**43 S.T.C.** estimated	6mm Villaboard™ over 10mm Standard Gypsum Wallboard
1 Hour 60/60/60	**45 S.T.C.**	6mm Villaboard™ over 13mm Fire Resistant Grade Gypsum Wallboard
1 1/2 Hours 90/90/ 90	**46 S.T.C.** estimated	6mm Villaboard™ over 16 mm Fire Resistant Grade Gypsum Wallboard
2 Hours 120/120/120	**48 S.T.C.**	9 mm Villaboard™ over 16 mm Fire Resistant Grade Gypsum Wallboard

All values are supported by Certificates and Letters of Opinion by independent Testing Authorities

Corner Detail

Two Hour Fire Rated Wall (Untiled)

Note: For tiled walls, refer to Fixing Diagrams opposite.



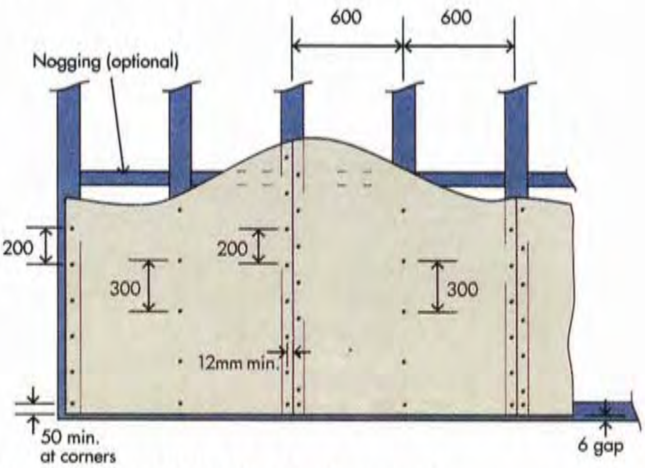
James Hardie Villaboard™
Drywall Partition Systems

Typical Fixing Diagrams

To be read in conjunction with James Hardie Villaboard™, Installation Manual

Vertically Fixed Villaboard™ (Non-Tiled Areas)

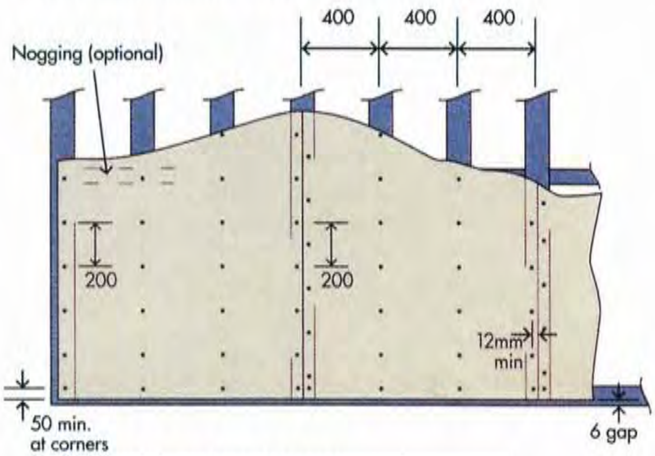
Typical for: 6mm, 9mm and 12mm Villaboard™



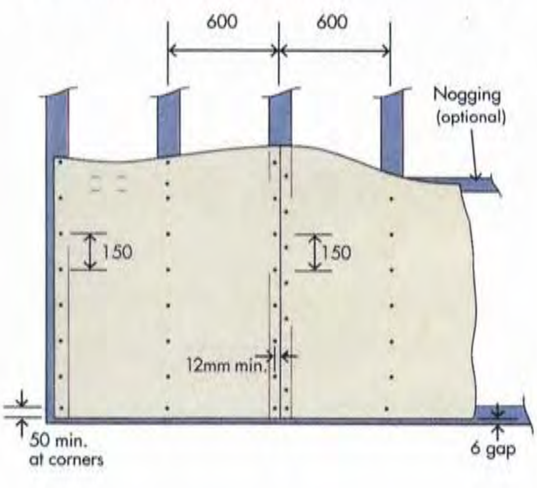
Vertically Fixed Villaboard™ (Tiled Areas)

Typical for: 6mm Villaboard™

For Stud Spacing: 400mm

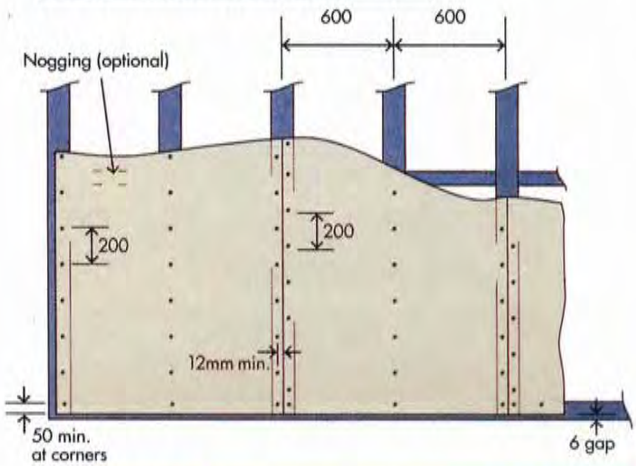


For Stud Spacing: 600mm



Typical for: 9mm and 12mm Villaboard™

For Stud Spacing: 600mm



NOTE:

- All dimensions are in millimetres (mm)
- Diagrams shown are typical for:
  - Villaboard™ Only Systems
  - Villaboard™ over Gypsum Wallboard Systems

James Hardie Villaboard™

INTERNAL DRYWALL SYSTEMS

WARRANTY

The systems recommended in this brochure are intended to assist those interested in the specification and installation of Villaboard™ drywall systems. However, the brochure is not intended to be an exhaustive statement of all relevant data. Further, as the successful installation of these systems depends on numerous factors outside the Company's control (e.g. quality of workmanship, particular design requirements, etc.), the Company accepts no responsibility for or in connection with the quality of the systems, or their suitability for any purpose when installed.

James Hardie & Coy Pty Limited warrants that its goods are free from defects caused by faulty manufacture or materials. If any of its goods are so defective, the Company will, at its option, either supply replacement goods or reimburse the buyer for the purchase price.

This warranty shall not apply to any loss or consequential loss suffered through or resulting from defects caused by faulty manufacture or materials.

All warranties other than those specified by the Company are hereby excluded, and all condition, obligations and liabilities however arising are hereby excluded. Nothing in this warranty, however, shall be construed as affecting any rights the buyer may have under any statute, rule or regulation or under the general law which gives the buyer rights which cannot be modified or excluded by agreement.

The contents of this brochure are subject to change without notice.

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Quality Endorsed Company AS 3902 Standards Australia

The StandardsMark is our customers' assurance that James Hardie & Coy Pty Limited will maintain their dedication to a quality management system and continual improvement in all aspects of the company's business. Fibre Cement Manufacturers: Sydney, Melbourne, Brisbane and Perth.

Installation Details

Contact your local James Hardie Distributor for all technical and sales enquiries.

Distributed By:

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